

Kalman Filter for Pairs Trading: Deep Dive

Study notes for the pairs-trading backtest project

1. Relation to regression

The Kalman filter is a regression that updates itself every day.

- **OLS:** one β for all data, every day weighted equally.
- **Rolling OLS:** refit on the last N days, a hard cutoff (day $N+1$ is fully ignored).
- **Kalman:** β is updated each day, and old data fades out smoothly. How fast it fades is set by one number (δ).

The special case where β is not allowed to move ($V_w = 0$) is recursive least squares: it gives the same answer as OLS, computed one day at a time. Adding the random-walk drift is what turns it into a forgetting regression.

2. Two equations are the whole model

- **State (what you can't see):** $\beta_t = \beta_{t-1} + \text{small random nudge}$, with nudge variance V_w .
- **Observation (what you can see):** $\log A_t = \alpha + \beta_t \cdot \log B_t + \text{noise}$, with noise variance V_e .

The filter's only job is to answer: given the prices so far, what is my best guess for β now, and how sure am I?

3. Two things the filter tracks

- **state:** the current best guess (α, β).
- **P:** how uncertain that guess is. This is the extra thing OLS doesn't have.

Everything depends on P . Uncertain means learn fast, confident means barely move.

4. The daily cycle, with the meaning of each line

1. **Predict the state:** tomorrow's β = today's β (random walk, so no trend assumed).
2. **Inflate uncertainty:** $R = P + V_w$. Time passed, so β might have moved, so you are less sure.
3. **Predict the observation:** $\hat{y} = \alpha + \beta \cdot \log B_t$.

4. **Surprise:** $e = \text{actual} - \hat{y}$.
5. **Expected surprise size:** $Q = x R x^T + V_e$. Part of it is your uncertainty about β , part is plain noise.
6. **Gain:** $K = R x^T / Q$. The share of the surprise you blame on the parameters rather than on noise.
7. **Update:** $\text{state} = \text{state} + K \cdot e$.
8. **Shrink uncertainty:** $P = R - K x R$. You learned something, so you are surer than before.

5. The gain K is the heart of it

Scalar version (one number to estimate, $x = 1$): $K = R / (R + V_e)$.

- **R large, V_e small:** K near 1. "I'm unsure and the data is clean, so I follow it."
- **R small, V_e large:** K near 0. "I'm confident and the data is noisy, so I mostly ignore it."

Worked example. You estimate a level $m = 10$, with $R = 4$ and $V_e = 4$. Then $K = 0.5$. Today's data says 12, so $e = 2$, and the new $m = 10 + 0.5 \cdot 2 = 11$. The new $P = R(1 - K) = 2$, so you are now more certain. Next day, with $V_w = 1$, $R = 2 + 1 = 3$ and $K = 3/7 \approx 0.43$. K shrinks as P shrinks, and that is learning.

6. What **delta** does

V_w is how fast you allow β to change.

- **Tiny V_w :** P collapses, K gets small, and the filter has long memory (nearly static OLS).
- **Large V_w :** P stays big, K stays big, and the filter chases every wiggle (short memory).

For small V_w , K settles around $\sqrt{V_w / V_e}$, and the memory is roughly $1/K$ days. So the choice of **delta** is a choice of how many days you trust the past, which is why it is fixed before looking at test results.

7. The lesson from the α bug

When α was allowed to random-walk at about 0.01 per day, it moved as fast as the spread itself, so the filter explained the spread away by shifting α . e shrank, z shrank, and there were no signals.

Rule: never let a parameter move as fast as the thing you are trading. That is why α is frozen and only β moves: β should drift slowly, and the spread should fluctuate fast.

8. Why $e / \sqrt{Q}$ is a z-score

e is the deviation of the spread from what the model expects, and $\sqrt{Q}$ is the model's own estimate of how large that deviation normally is. Their ratio is "how many standard deviations of surprise", with no rolling window needed.

9. Link to the paper

Equations (9) to (12) of Tenyakov and Mamon (2017) are the same filter in scalar form.

- The Kalman gain in Eq. (11), $M = \Sigma / (\Sigma + \alpha^2)$, is exactly $K = R / (R + V_e)$ from section 5.
- Eq. (12), $\Sigma_{k+1|k+1} = \Sigma_{k+1|k} - M \cdot \Sigma_{k+1|k}$, is the P update from step 8.
- The difference is what is hidden: in the paper the hidden state is the spread level r_k , and in this project it is β .

10. Where it can fail

- **V_e and V_w are assumptions**, not facts. Wrong values give wrong signals.
- It assumes the relationship changes smoothly. A sudden break (for example a merger) takes many days to be absorbed.
- It adapts to a changing β but cannot create mean reversion if the spread does not have any.

Self-check

If the observation noise V_e doubles and everything else stays the same, does K go up or down, and does β react faster or slower to a surprise?